

$$\begin{aligned}\mathbb{T}_{\mathcal{R}} &= \text{Hom}(F/F_{\text{norm}}, \mathcal{R}), \quad \dim \mathbb{T}_{\mathcal{R}} = 42, \quad A_{\tau} = A - \tau q_F D_{\alpha}|_{\mathcal{R}} \\ \mathcal{O}_{\text{tilt}}(A) &= \{B \in \text{End}(\mathcal{R}) : B(E) = -2E\}\end{aligned}$$

Choose a vector-space complement W : $\mathcal{R} = \mathbb{C}E \oplus W$, $W \cong \mathcal{R}/\mathbb{C}E$.

$$B = \begin{pmatrix} -2 & \ell \\ 0 & C \end{pmatrix}, \quad \ell \in \text{Hom}(W, \mathbb{C}E), \quad C \in \text{End}(W) \text{ arbitrary.}$$

The chosen complement need not be B -invariant.

Selected A

$$\chi_A(S) = (S + 2)^2 q(S)$$

$$B_0 = \text{diag}(-2, 0, \dots, 0)$$

$$\chi_{B_0}(S) = (S + 2)S^6$$

$$B_3 = \text{diag}(-2, 3, \dots, 3)$$

$$\chi_{B_3}(S) = (S + 2)(S - 3)^6$$

$$\gcd_{\tau} \chi_{A_{\tau}}(S) = \gcd_{\tau} m_{A_{\tau}}(S) = S + 2$$

The q -factor, second -2 eigenline and fixed g_3 -cyclic module are not invariants of this normalized internal tilt orbit.